

We make chemistry computable for AI.

242 validated tools across 18 domains in chemistry, analytics and process engineering, from molecular design to process capability, ready for any AI assistant.

21 % → 85 %

Public chemistry benchmark (MolecularQ): the same AI assistant without and with our tools, n = 3,540

242

Tools across 18 domains, in production since April 2026

18 vs. 125

Experiments for the same process understanding: computed design vs. grid search

1 Who we are

- **CovaSyn GmbH, Leipzig, Germany.** Founded 2025, in production since April 2026, entirely self-funded.
- **Two PhD chemists** with five years of pharma experience across CMC, drug discovery and GMP manufacturing. We build the software ourselves.

2 What we do for chemical producers

- **Fewer experiments for the same process understanding.** Design of experiments and optimisation return the factor ranking, the operating window and the process capability, not just a best guess.
- **Faster answers from the QC lab.** NMR, MS, IR, UV and HPLC data analysed in hours; impurities and hazards assessed, with the applicability domain stated.
- **AI assistants your teams can trust with numbers.** Our tools plug into the assistant you already run. Structure → solubility → experiment design → optimisation, chained with no manual handover.

3 What working with us looks like

- **A real run, not a demo.** Send us one molecule or one measurement series and you get the computed result back. No access needed, no meeting.
- **Then two routes.** Platform access for your teams and your own systems, or a joint project: you do chemistry and testing, we do the data side.
- **What to expect.** Results in hours instead of weeks, documented and repeatable, and a clear statement of where we are not the right tool.

On the benchmark figure: the yardstick is not ours (MolecularQ, ICLR 2026, JKU Linz), the measurement is. Details and our known weaknesses on page 2.

Measured, not claimed.

Every result on these pages comes from real runs of our production system. Where we fall short, we say so.

THE BENCHMARK

MolecularQ chemistry questions	AI assistant alone	with CovaSyn tools		questions
Fast AI assistant	21 %	85 %		3,540
Frontier AI assistant	41 %	92 %		910

Both assistants from the same provider. Disclosed up front: the largest remaining error is the assistant overriding a correct tool result, and in 1 of 64 question categories our tools made the answer worse. The cause is identified.

WORKED EXAMPLE: ONE PROCESS DATASET, THREE METHODS, ONE ANSWER

STEP 1 · 11 RUNS

Screening

Temperature dominates, then catalyst loading, then time. The base has negligible effect and is dropped.

Curvature detected: a straight-line model is not enough.



STEP 2 · 18 RUNS

Response surface

R^2 **0.92**. Optimum 84.7 °C · 1.39 mol% · 11.1 h → **93.6 %** yield (CI 93.0 to 94.3).

Proven acceptable range for yield $\geq 92\%$.



STEP 3 · SAME 29 RUNS

Machine learning

Q^2 **0.89** for yield; temperature again ranked first. Set point 85.2 °C → **93.4 %**.

Purity flagged **monitor only** (Q^2 0.53), not oversold.

Illustrative catalytic coupling dataset; every figure is a verbatim tool output. The three answers agree because the data agree, not because we aligned them.

WHERE WE STAND, SEPTEMBER 2026

- **In production** since 16 April 2026, as a cloud service with no installation.
- **242 tools, 18 domains** for any AI assistant, through the openModel Context Protocol.
- **Enterprise-ready**: company single sign-on and a contract set for pilots.
- **Three collaborations started** through expert contact: a Leibniz institute, a Fraunhofer institute and a biotech company.

WHAT WE CANNOT DO YET

- **Nopaying customers** on the platform yet. Self-funded to date.
- **Cloud only**. Delivery behind the customer firewall is designed and planned for 2028.
- **Our strength is fine and specialty chemicals, pharma and biologics**. No polymer-specific models, no plant-scale reactor simulation, no advanced process control.

NEXT MILESTONES

Q4 2026

Paid subscriptions live; first signed research-institute agreement

Q1 2027

External validation of selected predictions on partner measurement data

2027

First sales hire; first pilot outside the German-speaking market

2028

On-premise delivery; operating break-even (plan)